GGG-W-00686C(GSA-FSS)
Amendment-1
December 10, 1971

AMENDMENT

TO

INTERIM FEDERAL SPECIFICATION

WRENCH, TORQUE

This amendment was developed by the Standardization Division, Federal Supply, Service, General Services Administration, Washington, DC 20406, based upon currently available technical information. The General Services Administration has authorized this amendment as a part of Interim Federal Specification GGG-W-00686C(GSA-FSS), dated October 15, 1969.

PAGE 1

Paragraph 1.2.1, under "Type I": Delete "Defecting" and substitute "Deflecting".

PAGE 2

Paragraph 2.1, under "Federal Specifications:" Delete the following:

"QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For.
TT-E-489 - Enamel Alkyd, Gloss (for Exterior and Interior Surfaces)."

Paragraph 2.1, under "Federal Specifications:" Add "PPP-P-40 - Packaging and Packing of Hand Tools."

Paragraph 2.1, under "Federal Standards:" Delete "Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)."

Paragraph 2.1, "Military Specifications:" Delete in its entirety.

Paragraph 2.1, under "Military Standards:" Delete "MIL-STD-129 - Marking for Shipment and Storage."

FSC 5120

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PAGE 3

Add the following new paragraph:

"3.2 Preproduction sample. Two wrenches of each type, class, style, and size shall be submitted for testing in accordance with the requirements of 4.7."

Paragraph 3.2: Renumber as "3.3".

Paragraph 3.3: Renumber as "3.3.1".

Paragraph 3.6, first sentence: After "wrenches" insert "that have torque presetting devices".

Paragraph 3.7, first sentence: Delete and substitute: "The torque wrenches shall be provided with a square drive with spring loaded steel ball in accordance with the dimensional requirements of GGG-W-641, except hardness shall be not less than Rockwell C 34 for wrenches with torque capacity ranges in inch-pounds and foot-pounds. Wrenches with torque capacity ranges in inch-grams and inch-ounces are not required to have their tangs hardened if all other applicable requirements of this specification are met."

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Paragraph 3.8.1, last sentence: Delete and substitute: "The width or any graduation line shall be less than 25 percent of the adjacent space width."

Paragraph 3.8.3, fourth line: Change "0.030" to "0.060".

Paragraph 3.8.4, first sentence: Delete "---shall not exceed the narrowest graduation line width plus 25 percent" and substitute "---shall not exceed 0.015 inch or the narrowest graduation line width plus 25 percent".

Paragraph 3.8.4, last sentence: Delete and substitute: "The pointer shall not cover the entire length of the graduation line unless that part of the pointer covering the graduation lines is no greater in width than the narrowest graduation line width plus 25 percent."

Paragraph 3.11, second sentence: After the word "Wrenches" add "less than 1,000 foot-pound capacity".

Paragraph 3.11: Add to paragraph: "(See footnote table XI and 4.8.)"

Paragraph 3.12.1, second sentence: Delete and substitute: "The pointers of the types I and II wrenches shall return to zero after use and the zero position of the pointer shall remain within 1/4 increment (see table II) in an unloaded condition regardless of the position of the wrench (see 4.5)."

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TABLE I: Delete and substitute:

TABLE I. Wrench tolerances

Percent of full scale or capacity		Tolerance plus or minus percent of reading			Type IV
From	Including	Types I and II	Type III		
			clockwise	counterclockwise	
20*	Full scale	4 percent	4 percent	6 percent	Tolerance = 5% of setting, or, as specified below whichever is greater: Size Tolerances 1 ±1/2 inch ounce 2 ±1/2 inch pound

* No accuracy requirement from 0 percent thru 20 percent of full scale or capacity of wrench.

Paragraphs 3.13 through 3.13.2: Delete in their entirety and substitute:

"3.13 Finish and coatings. All external wrench parts shall be smoothly finished. External ferrous parts, except the male tang and ball, the bezels and indicator plates, external locking devices, screws and other fasteners, shall be protected with a chromium plated coating. All external parts shall be protected to resist corrosion. Internal parts subject to corrosion shall be protected by the use of oil or other protective coating. Chromium plating shall be applied in accordance with QQ-C-320, class 1. Cadmium plating shall not be used as a protective coating."

Paragraph 3.14: Delete and substitute:

"3.14 Marking. Each wrench shall be marked in a legible and permanent manner with the manufacturer's name and model or part number. Type III wrenches shall be permanently marked with the word "TORQUES" and marked with one or two arrows, as applicable, to indicate direction. Wrenches shall also be permanently marked with the month and year of manufacture (e.g., 10-69 for October 1969)."

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Paragraph 3.15: Insert after second sentence: "Cases for type III wrenches shall be of sufficient length to accomodate the wrench in its unloaded condition (i.e., wrench set at its minimum setting)."

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Paragraph 3.15, line 3: Delete "suitable catch" and substitute "which can not be opened without previous disengagement of a positive latch,".

Paragraph 3.15: Delete data at end of sentence and substitute:

"Wrench, Torque Capacity _____.
Manufacturer's name and model or part number _____.
FSN _____."

Paragraph 3.16: Add to paragraph: "Type I wrenches shall have a pivoted type handle or other suitable design to assure accurate results."

Paragraph 3.16.2: Delete second sentence.

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Paragraph 3.17: Add to paragraph: "The handle design must provide a means for fixing the lever length of the torque wrench to provide an accurate means for the correct use to torque wrench adapter combinations."

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Paragraph 3.18.1: Delete and substitute:

"3.18.1 Accuracy calibration and sealing. Type III wrenches shall be correctly calibrated at the manufacturer's plant in accordance with the accuracy requirements of 3.12.1, after which they shall be protected against tampering by sealing the adjusting mechanism with a fusible alloy of low melting point."

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Paragraph 3.20: Delete and substitute:

"3.20 Instructions. The manufacturer shall furnish one copy of operating and maintenance instructions with each wrench. The instructions shall instruct the user concerning the use of adapters. The instructions shall indicate that no accuracy guarantee exists in the 0 to 20 percent of full load or capacity range."

TABLE IX, last column: Delete "4.5" and substitute "4.8".

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Paragraph 4.4: Delete and substitute:

"4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 thru 4.4.3.4 and 4.8."

Paragraph 4.4.2, last sentence: Delete and substitute: "All test load positions shall be approached from the low side without exceeding the target load. If the target load is exceeded, the test procedure must be repeated. For type I wrenches, the load shall be applied on the pivot pin in a direction tangential to the arc of the load applied. For type III wrenches, the load shall be applied smoothly at a rate not in excess of one percent of wrench capacity per second as the wrench approaches the time the signal mechanism actuates."

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TABLE XI, footnote **: Delete "---lowest calibrated scale reading and---".

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Paragraph 4.4.4: Delete and substitute:

"4.4.4 Hardness test for drive squares. Each preproduction sample wrench shall be subjected to a hardness value for drive squares as specified in 3.7. Test shall be in accordance with Method 243.1 of Fed. Std. No. 151."

Paragraph 4.4.5: Delete and substitute:

"4.4.5 Handgrips. The handgrip of one of the preproduction sample wrenches shall be subjected to the tests of 4.4.5.1 and 4.4.5.2, if applicable. If the wrench fails, the other preproduction sample wrench shall be tested. Failure of the second wrench to pass the tests shall be cause for rejection."

Paragraph 4.4.5.1: Delete and substitute:

"4.4.5.1 Handgrip shatter test. The handgrip shall be placed on a foundation and a suitable guard to protect the operator of the test shall be placed around the handgrip. A steel weight of 5 pounds shall be dropped with a flat surface directly upon the handgrip from a 3-foot height, two times. If the handgrip is functional and unbroken after the two direct blows, it shall have passed the shatter test."

Paragraph 4.4.5.2: Add to title "(handgrips other than metal or wood)."

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Paragraph 4.5: Delete and substitute:

"4.5 Endurance tests for preproduction samples. Endurance tests shall be performed on one sample only. Ratcheting mechanisms need not comply with the requirements of the cycling test, but shall be tested in accordance with GGG-W-641. Fixed heads may be substituted for ratchet heads to perform the required cycle tests.

Paragraphs 4.5.1 and 4.5.2: Delete and substitute:

"4.5.1 Types I, II, and III wrenches. Each sample wrench shall be stressed from a no-load condition to a full load condition 5,000 times in a clockwise direction and 5,000 times in a counterclockwise direction. Type III, classes 1 and 2 wrenches shall be stressed from a no-load condition to a full load condition 10,000 times in a clockwise direction. Type I wrenches of less than 2,000 foot-pound capacity shall be stressed at a rate of not more than 120 times per minute; wrenches of 2,000 foot-pound capacity and over shall be stressed at a rate of not more than 10 times per minute. Types II and III wrenches of less than 2,000 foot-pound capacity shall be stressed at a rate of not more than 20 times per minute; wrenches of 2,000 foot-pound capacity and over shall be stressed at a rate of not more than 10 times per minute. At the conclusion of this test, without performance of maintenance or recalibration, the wrenches shall be subjected to the test of 4.4.3. If the wrench passes the test of 4.4.3, it shall then be stressed from a no-load condition to a full load condition 10,000 times in a clockwise direction and 10,000 times in a counterclockwise direction. Type III, classes 1 and 2 shall be stressed from a no-load condition to a full load condition 20,000 times in a clockwise direction. Accurate performance shall not be required at the conclusion of this test, but deformation or physical failure shall be cause for rejection. After minor maintenance and recalibration, if required, the wrench shall be subjected to the test of 4.4.3."

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Paragraph 4.5.3: Delete and substitute:

"4.5.2 Type IV wrenches. Type IV wrenches shall be stressed from a no-load condition to the maximum rated load 50,000 times at a rate not more than 20 times per minute. At the conclusion of the test, the wrench shall be subjected to the test of 4.4.3.4."

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Add the following new paragraphs:

"4.7 Preproduction samples. Each preproduction sample unit shall be inspected and tested in accordance with 4.3 and 4.4 prior to regular production on a contract. The tests shall be witnessed by a Government representative and failure of any test shall be cause for rejection of the preproduction samples. Wrenches having a drive size smaller than normal for a given capacity (see applicable tables) may have an identical wrench of the same capacity but with a larger size drive tang substituted for it in the endurance tests of 4.5, for all other test requirements, the smaller drive size as specified for the wrench shall be used.

4.8 Endurance tests for lot acceptance. For lot acceptance, the wrenches shall be cycled at the rates specified for the type wrench being tested, with the total number of cycles at full load for each wrench: Types I, II, and III, classes 3 and 4; 1,000 times in each direction, type III, classes 1 and 2, and type IV; 2,000 times in the direction the wrench indicates torque. Without recalibration, the wrenches shall be subjected to the test of 4.4.3. Wrenches with a capacity exceeding the sizes specified in table XI shall be tested in accordance with 3.11."

Paragraphs 5.1 thru 5.2.2: Delete these paragraphs and insert the following:

"5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or C, as specified (see 6.2)."

Paragraph 6.2, (j): Delete this subparagraph.

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